

BRINEX SYSTEM

Performance Test Report

Microservices Load Testing - Grafana k6

Test Date: March 7–8, 2026

Test Tool: Grafana k6

Services Tested: 5 Core Microservices

Test Types: Smoke · Average Load · Stress

1. Executive Summary

This report presents the results of performance testing conducted on the Brinex System microservices architecture. Testing was performed using Grafana k6 on March 7–8, 2026 and covered three test scenarios - Smoke, Average Load, and Stress - across five core services: User, Compass, Crystallization, Vision, and Waste Valorization.

The Smoke test validates baseline functionality under minimal load (1 VU). The Average Load test simulates normal operating conditions (5 VUs). The Stress test evaluates system behaviour under peak load (15 VUs).

1.1 Overall Results at a Glance

Service	Smoke	Avg Load	Stress	Peak VUs	Overall
User Service	PASS	PASS	PASS	15 VUs	PASS
Compass Service	PASS	PASS	WARN	15 VUs	WARN
Crystallization Service	PASS	PASS	PASS	15 VUs	PASS
Vision Service	PASS	WARN	WARN	15 VUs	WARN
Waste Valorization Service	PASS	PASS	PASS	15 VUs	PASS

Key findings: Crystallization and Waste Valorization services demonstrate strong, production-ready performance. User Service is stable with minor tuning recommended. Compass and Vision services exhibit significant degradation under load and require optimization before production deployment.

2. Test Configuration

2.1 Test Scenarios

Test Type	Virtual Users	Duration	Purpose	Pass Criteria
Smoke Test	1 VU	~30 sec	Validate core functionality with minimal load	0% errors, all checks pass
Average Load Test	5 VUs	~2 min	Simulate normal daily usage conditions	Error rate <50%, checks >95%
Stress Test	15 VUs	~3.5 min	Identify breaking points and degradation	System stable, no crashes

2.2 Services Under Test

Service	Endpoints Tested	Description
User Service	Get By Email, Get By ID, Get All	Core user management and authentication service
Compass Service	Deals, Offers, Harvest Plans	Business intelligence and planning data service
Crystallization Service	Salt Production, Model Performance, Daily Measurement, Predicted Daily/Monthly	ML prediction and crystallization analytics service
Vision Service	Health, Batches, Stats, Detections	Computer vision and image detection service
Waste Valorization Service	Jobs, Job Status, Predictions, Quick Prediction	Waste processing job management and prediction service

3. Smoke Test - All Services

The Smoke Test runs with a single Virtual User (1 VU) to confirm that all service endpoints are reachable, return correct responses, and pass functional checks under zero load. A failed smoke test indicates a fundamental issue that must be resolved before any load testing.

Service	Avg (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Req/s	Checks	Status
User Service	299.31 ms	178.83 ms	549.66 ms	576.71 ms	732.74 ms	1.034	100%	PASS
Compass Service	212.78 ms	4.87 ms	544.57 ms	1020.53 ms	2287.53 ms	1.309	98.75%	PASS
Crystallization Service	114.76 ms	84 ms	151.15 ms	154.42 ms	886.08 ms	1.475	100%	PASS
Vision Service	468.64 ms	340.81 ms	1102.76 ms	1151.12 ms	1350.95 ms	0.961	100%	PASS
Waste Valorization Service	3.56 ms	3.42 ms	4.55 ms	5.66 ms	6.54 ms	1.655	100%	PASS

Key Observations

- Crystallization and Waste Valorization services achieved perfect 100% check rates.
- Compass Service had 1 check failure (98.75%) and a high HTTP failure rate (70%) driven by expected invalid-state requests.
- Vision Service performed adequately at single VU with no failures.
- User Service had a 33.33% HTTP failure rate (within threshold) with all checks passing.

4. Average Load Test - All Services

The Average Load Test simulates 5 concurrent Virtual Users operating over a sustained period, representing expected normal daily traffic. This test identifies how services perform under realistic production conditions and whether response times remain acceptable.

Service	Avg (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Req/s	Checks	Status
User Service	235.22 ms	165.68 ms	371.37 ms	413.92 ms	456.48 ms	1.086	100%	PASS
Compass Service	1160.61 ms	4.19 ms	968.1 ms	1418.51 ms	60000.29 ms	2.12	98.11%	PASS
Crystallization Service	92.15 ms	76.39 ms	142.91 ms	153.24 ms	693.16 ms	6.019	100%	PASS
Vision Service	3151.37 ms	1439.94 ms	9017.54 ms	12343.61 ms	36436.28 ms	1.113	89.17%	WARN
Waste Valorization Service	3.6 ms	3.37 ms	4.45 ms	4.91 ms	28.28 ms	6.457	100%	PASS

Key Observations

- Crystallization Service maintained 100% checks and sub-100ms average response times - outstanding.
- Waste Valorization maintained sub-5ms response times with 100% checks across 780 requests.
- Vision Service degraded significantly: 3,151 ms average and 89.17% check rate - below acceptable threshold.
- Compass Service showed extreme degradation with a 60,000 ms maximum (timeout) and 98.11% checks.
- User Service remained stable at 235 ms average with 100% checks under 5 VUs.

5. Stress Test - All Services

The Stress Test pushes each service to 15 concurrent Virtual Users to identify performance limits, degradation patterns, and potential breaking points. High response times, failed checks, or timeouts under stress indicate areas requiring optimization before handling peak production traffic.

Service	Avg (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Req/s	Checks	Status
User Service	374.74 ms	226.09 ms	633.29 ms	759.86 ms	11811.39 ms	9.137	98.71%	PASS
Compass Service	2376.35 ms	4.62 ms	2748.18 ms	12796.64 ms	60001.26 ms	3.317	93.45%	WARN
Crystallization Service	88.64 ms	76.65 ms	134.45 ms	146.66 ms	703.34 ms	14.434	100%	PASS
Vision Service	6534.54 ms	1712.39 ms	20048.59 ms	25970.79 ms	60003.15 ms	1.267	84.17%	FAIL
Waste Valorization Service	11.01 ms	4.32 ms	18.27 ms	47.09 ms	335.23 ms	15.594	100%	PASS

Key Observations

- Crystallization Service excelled: 88.64 ms average response, 14.43 req/s, 100% checks - best overall performance.
- Waste Valorization maintained 11 ms average even at 15 VUs processing 3,285 requests - highly efficient.
- User Service held at 374 ms average with 98.71% checks - acceptable degradation.
- Vision Service hit 60,003 ms timeouts on Stats endpoint; check rate fell to 84.17% - critical failure zone.
- Compass Service shows 71.13% HTTP failure rate (still within threshold) but with 93 check failures - needs urgent attention.

6. Service Deep Dive Analysis

This section provides a detailed breakdown of each service's performance across all three test types, including per-endpoint response time statistics and targeted recommendations.

6.1 User Service

Test / Endpoint	Avg (ms)	Min (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Checks %	Status
Smoke Test - VUs: 1 Duration: 31.9s Iterations: 11 Total Requests: 33								
Overall HTTP	299.31 ms	138.32 ms	178.83 ms	549.66 ms	576.71 ms	732.74 ms	100%	PASS
User Get By Email	179.85 ms	138.32 ms	159.68 ms	253.86 ms	278.17 ms	302.48 ms	-	-
User Get All	534.39 ms	426.29 ms	530.93 ms	608.14 ms	670.44 ms	732.74 ms	-	-
User Get By Id	183.7 ms	143.73 ms	165.05 ms	260.19 ms	287.38 ms	314.56 ms	-	-
Average Load Test - VUs: 5 Duration: 3.7s Iterations: 1 Total Requests: 4								
Overall HTTP	235.22 ms	153.03 ms	165.68 ms	371.37 ms	413.92 ms	456.48 ms	100%	PASS
User Get By Id	172.8 ms	172.8 ms	172.8 ms	172.8 ms	172.8 ms	172.8 ms	-	-
User Get All	456.48 ms	456.48 ms	456.48 ms	456.48 ms	456.48 ms	456.48 ms	-	-
User Get By Email	155.8 ms	153.03 ms	155.8 ms	158.01 ms	158.29 ms	158.56 ms	-	-
Stress Test - VUs: 15 Duration: 212.1s Iterations: 646 Total Requests: 1938								
Overall HTTP	374.74 ms	133.53 ms	226.09 ms	633.29 ms	759.86 ms	11811.39 ms	98.71%	PASS
User Get By Email	222.66 ms	133.53 ms	176.08 ms	284.52 ms	352.19 ms	2017.89 ms	-	-
User Get All	686.95 ms	419.24 ms	516 ms	773.29 ms	1134.33 ms	11811.39 ms	-	-
User Get By Id	214.62 ms	134.99 ms	172.41 ms	312.28 ms	331.11 ms	1459.5 ms	-	-

Findings

- Smoke test passed all threshold checks (100% check rate) with a single VU handling 33 requests across 3 endpoints.
- The "Get All Users" endpoint is the slowest operation, averaging 534 ms (smoke) and spiking to 11,811 ms max under stress - this is the primary bottleneck.
- Stress test with 15 VUs generated 1,938 requests at 9.1 req/s with a 98.71% check rate - largely stable.
- HTTP failure rate of 33.33% is consistent across all test types but the threshold is set to <50%, meaning it passes. This rate likely reflects expected 404/auth rejections in the test scenario rather than true errors.

Recommendations

- Optimize the Get All Users query - add pagination or caching (Redis) to avoid full-table scans.
- Investigate the 11.8s spike in stress test; likely a database connection pool exhaustion.
- Review if the 33% HTTP failure rate is intentional (expected rejections) or needs threshold tightening.

6.2 Compass Service

Test / Endpoint	Avg (ms)	Min (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Checks %	Status
Smoke Test - VUs: 1 Duration: 30.6s Iterations: 4 Total Requests: 40								
Overall HTTP	212.78 ms	2.89 ms	4.87 ms	544.57 ms	1020.53 ms	2287.53 ms	98.75%	PASS
Compass Deals	331.1 ms	2.89 ms	4.4 ms	1037.44 ms	1364.05 ms	2287.53 ms	-	-
Compass Harvest Plans	104.16 ms	3 ms	5.51 ms	320.78 ms	404.84 ms	495.75 ms	-	-
Compass Offers	163.64 ms	3.03 ms	5.1 ms	498.49 ms	505.29 ms	507.58 ms	-	-
Average Load Test - VUs: 5 Duration: 150s Iterations: 30 Total Requests: 318								
Overall HTTP	1160.61 ms	0.73 ms	4.19 ms	968.1 ms	1418.51 ms	60000.29 ms	98.11%	PASS
Compass Harvest Plans	266.36 ms	2.4 ms	4.06 ms	274.17 ms	645.35 ms	8732.05 ms	-	-
Compass Deals	1150.9 ms	0.88 ms	4.3 ms	1273.75 ms	1821.59 ms	49310.63 ms	-	-
Compass Offers	2087.1 ms	0.73 ms	4.12 ms	697.41 ms	1093.07 ms	60000.29 ms	-	-
Stress Test - VUs: 15 Duration: 214.1s Iterations: 71 Total Requests: 710								
Overall HTTP	2376.35 ms	0.57 ms	4.62 ms	2748.18 ms	12796.64 ms	60001.26 ms	93.45%	WARN
Compass Offers	2498.19 ms	0.7 ms	5.06 ms	3644.42 ms	28826.82 ms	60000.43 ms	-	-
Compass Deals	2887.59 ms	0.57 ms	4.42 ms	2980.43 ms	13354.09 ms	60001.26 ms	-	-
Compass Harvest Plans	1572.85 ms	1.97 ms	5.1 ms	1743.63 ms	8885.52 ms	41179.76 ms	-	-

Findings

- Compass Service shows critical performance degradation under load - average response time escalates from 213 ms (smoke) to 1,161 ms (average load) to 2,376 ms (stress).
- Maximum response times hit 60,001 ms (60 seconds - timeout boundary) across average and stress tests, indicating severe bottlenecks.
- Check rate degrades from 98.75% (smoke) to 93.45% (stress), with 93 failed checks under stress - the most failures of any service.
- The Compass Offers endpoint is the worst performer: 2,498 ms avg and 60,000 ms max under stress.
- Compass Deals endpoint reaches 2,888 ms avg and 60,001 ms max under stress - near timeout conditions.

Recommendations

- **CRITICAL:** Compass Service requires immediate optimization - 60s timeouts under average load are unacceptable for production.
- Introduce database query optimization, connection pooling, and consider read replicas for Deals/Offer endpoints.
- Implement response caching for harvest plans and offers data which are likely read-heavy.
- Set up circuit breakers to prevent cascade failures when response times approach timeout thresholds.

6.3 Crystallization Service

Test / Endpoint	Avg (ms)	Min (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Checks %	Status
Smoke Test - VUs: 1 Duration: 32.5s Iterations: 6 Total Requests: 48								
Overall HTTP	114.76 ms	70.68 ms	84 ms	151.15 ms	154.42 ms	886.08 ms	100%	PASS
Cryst Predicted Monthly	77.75 ms	71.82 ms	76.88 ms	83.17 ms	84.93 ms	86.69 ms	-	-
Cryst Model Performance	82.5 ms	74.41 ms	79.35 ms	92.95 ms	93.36 ms	93.76 ms	-	-
Cryst Daily Measurement	183.9 ms	71.72 ms	141.28 ms	161.81 ms	488.17 ms	886.08 ms	-	-
Cryst Predicted Daily	143.55 ms	133.44 ms	141.32 ms	153.62 ms	153.83 ms	154.04 ms	-	-
Cryst Salt Production	82.15 ms	70.68 ms	78.8 ms	93.93 ms	104.5 ms	116.74 ms	-	-
Average Load Test - VUs: 5 Duration: 121s Iterations: 91 Total Requests: 728								
Overall HTTP	92.15 ms	61.02 ms	76.39 ms	142.91 ms	153.24 ms	693.16 ms	100%	PASS
Cryst Salt Production	80.65 ms	65.08 ms	74.23 ms	92.39 ms	109.74 ms	282.25 ms	-	-
Cryst Predicted Monthly	77.61 ms	67.91 ms	74.84 ms	87.29 ms	92.59 ms	198.92 ms	-	-
Cryst Predicted Daily	121.37 ms	70.98 ms	135.73 ms	153.84 ms	167.84 ms	338.16 ms	-	-
Cryst Daily Measurement	108.09 ms	61.02 ms	79.84 ms	150.64 ms	173.44 ms	693.16 ms	-	-
Cryst Model Performance	80.1 ms	69.14 ms	75.43 ms	88.89 ms	95.29 ms	200.06 ms	-	-
Stress Test - VUs: 15 Duration: 215.6s Iterations: 389 Total Requests: 3112								
Overall HTTP	88.64 ms	61.12 ms	76.65 ms	134.45 ms	146.66 ms	703.34 ms	100%	PASS
Cryst Daily Measurement	96.15 ms	67.52 ms	79.34 ms	145.22 ms	158.37 ms	654.1 ms	-	-
Cryst Predicted Daily	113.12 ms	69.72 ms	91.9 ms	156.57 ms	181.77 ms	703.34 ms	-	-
Cryst Salt Production	80.44 ms	61.12 ms	74.57 ms	94.85 ms	110.68 ms	650.65 ms	-	-
Cryst Model Performance	79.99 ms	62.57 ms	75.37 ms	95.28 ms	109.07 ms	161.43 ms	-	-
Cryst Predicted Monthly	82.35 ms	61.88 ms	74.97 ms	94.05 ms	108.4 ms	670.5 ms	-	-

Findings

- Crystallization Service is the strongest performer overall - 100% check rates across all three test types with zero threshold failures.
- Average response time actually improves slightly from 114.76 ms (smoke) to 92.15 ms (average) to 88.64 ms (stress), demonstrating excellent scaling behavior.
- Stress test processed 3,112 requests at 14.43 req/s - the highest throughput of any service tested.
- All 5 endpoints (salt production, model performance, daily measurement, predicted daily/monthly) remain well within acceptable response times even at 15 VUs.
- Maximum observed response time of 703 ms under stress is well controlled relative to other services.

Recommendations

- No immediate action required - Crystallization Service is production-ready.
- Consider this service architecture as a reference implementation for other services.
- Monitor the Daily Measurement endpoint which showed a one-time 886 ms spike in smoke test.

6.4 Vision Service

Test / Endpoint	Avg (ms)	Min (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Checks %	Status
Smoke Test - VUs: 1 Duration: 36.4s Iterations: 5 Total Requests: 35								
Overall HTTP	468.64 ms	4.9 ms	340.81 ms	1102.76 ms	1151.12 ms	1350.95 ms	100%	PASS
Vision Batches	253.23 ms	149.85 ms	167.77 ms	410.94 ms	441.28 ms	471.62 ms	-	-
Vision Health	7.79 ms	4.9 ms	7.24 ms	10.05 ms	10.39 ms	10.72 ms	-	-
Vision Stats	700.36 ms	163.23 ms	641.27 ms	1149.08 ms	1167.88 ms	1350.95 ms	-	-
Vision Detections	218 ms	149.13 ms	209.94 ms	287.4 ms	298.32 ms	309.24 ms	-	-
Average Load Test - VUs: 5 Duration: 125.8s Iterations: 20 Total Requests: 140								
Overall HTTP	3151.37 ms	4.27 ms	1439.94 ms	9017.54 ms	12343.61 ms	36436.28 ms	89.17%	WARN
Vision Detections	6100.42 ms	228.51 ms	2090.51 ms	15365.91 ms	29311.26 ms	36436.28 ms	-	-
Vision Stats	3553.16 ms	145.99 ms	1824.33 ms	10350.18 ms	12343.61 ms	22073.88 ms	-	-
Vision Batches	1741.15 ms	180.4 ms	1216.92 ms	3595.01 ms	6270.18 ms	7445.72 ms	-	-
Vision Health	5.38 ms	4.27 ms	5.23 ms	6.38 ms	6.7 ms	6.89 ms	-	-
Stress Test - VUs: 15 Duration: 240s Iterations: 36 Total Requests: 304								
Overall HTTP	6534.54 ms	3.69 ms	1712.39 ms	20048.59 ms	25970.79 ms	60003.15 ms	84.17%	FAIL
Vision Stats	8699.89 ms	149.62 ms	3724.14 ms	23514.78 ms	26171.83 ms	60003.15 ms	-	-
Vision Health	5.85 ms	3.69 ms	5.14 ms	8.15 ms	8.53 ms	11.44 ms	-	-
Vision Batches	4953.6 ms	145.7 ms	2363.43 ms	10018.85 ms	11251.89 ms	42873.69 ms	-	-
Vision Detections	7134.52 ms	148.42 ms	2130.51 ms	13237.55 ms	38963.08 ms	53310.55 ms	-	-

Findings

- Vision Service handles single-VU smoke tests well (0% failure, 100% checks) but degrades significantly under concurrent load.
- Average response time jumps from 469 ms (smoke) to 3,151 ms (average load) to 6,535 ms (stress) - a 14x increase from smoke to stress.
- Stress test check rate drops to 84.17% with 85 failed checks - the second worst degradation after Compass.
- The Vision Stats endpoint is the most resource-intensive: 8,700 ms avg and 60,003 ms max under stress (timeout reached).
- Vision Detections endpoint reaches 7,135 ms avg and 53,311 ms max under stress - indicating heavy ML compute overhead.

- The health endpoint remains consistently fast (~5-8ms) confirming the service itself is alive but compute-bound.

Recommendations

- CRITICAL: Vision Service is not production-ready under concurrent load - requires significant optimization.
- Implement asynchronous processing for detection and stats computations with a job queue (e.g., Kafka, BullMQ).
- Add GPU acceleration or model optimization (quantization, ONNX) to reduce ML inference time.
- Implement request queuing and rate limiting to prevent resource starvation under concurrent access.
- Consider horizontal scaling (multiple inference workers) for the detection and stats endpoints.

6.5 Waste Valorization Service

Test / Endpoint	Avg (ms)	Min (ms)	Med (ms)	P90 (ms)	P95 (ms)	Max (ms)	Checks %	Status
Smoke Test - VUs: 1 Duration: 30.2s Iterations: 10 Total Requests: 50								
Overall HTTP	3.56 ms	0.8 ms	3.42 ms	4.55 ms	5.66 ms	6.54 ms	100%	PASS
Waste Job Status	3.51 ms	2.68 ms	3.36 ms	4.01 ms	4.53 ms	5.06 ms	-	-
Waste Quick Prediction	3.48 ms	0.8 ms	3.49 ms	4.38 ms	4.7 ms	5.03 ms	-	-
Waste Jobs	3.5 ms	0.83 ms	3.43 ms	4.33 ms	6.18 ms	6.54 ms	-	-
Waste Predictions	3.8 ms	2.7 ms	3.54 ms	4.67 ms	5.41 ms	6.16 ms	-	-
Average Load Test - VUs: 5 Duration: 120.8s Iterations: 156 Total Requests: 780								
Overall HTTP	3.6 ms	1.51 ms	3.37 ms	4.45 ms	4.91 ms	28.28 ms	100%	PASS
Waste Quick Prediction	3.64 ms	1.81 ms	3.33 ms	4.53 ms	4.86 ms	16.46 ms	-	-
Waste Jobs	3.57 ms	1.91 ms	3.33 ms	4.46 ms	5.01 ms	14.9 ms	-	-
Waste Job Status	3.82 ms	1.51 ms	3.53 ms	4.43 ms	4.8 ms	28.28 ms	-	-
Waste Predictions	3.4 ms	1.71 ms	3.28 ms	4.34 ms	4.66 ms	5.47 ms	-	-
Stress Test - VUs: 15 Duration: 210.7s Iterations: 657 Total Requests: 3285								
Overall HTTP	11.01 ms	0.95 ms	4.32 ms	18.27 ms	47.09 ms	335.23 ms	100%	PASS
Waste Predictions	11.8 ms	1.76 ms	4.34 ms	15.44 ms	47.48 ms	335.23 ms	-	-
Waste Quick Prediction	9.32 ms	2.08 ms	4.31 ms	14.26 ms	34.68 ms	196.79 ms	-	-
Waste Jobs	11.23 ms	0.95 ms	4.3 ms	20.45 ms	49.93 ms	268.92 ms	-	-
Waste Job Status	11.45 ms	2.09 ms	4.35 ms	19.52 ms	51.23 ms	180.47 ms	-	-

Findings

- Waste Valorization Service shows exceptionally fast response times - averaging 3.56 ms (smoke), 3.60 ms (average), and 11.01 ms (stress).
- All endpoints consistently respond in single-digit milliseconds, suggesting responses may be cached or the service hits in-memory data.
- 100% check rates across all test types with zero check failures.
- Stress test delivered 3,285 requests at 15.59 req/s - the highest request rate of all services.
- Note: The http_req_failed threshold was not defined in test scripts (null), so pass/fail verdict is based on check rates alone.
- Maximum response time of 335 ms under stress is the lowest ceiling of all services tested.

Recommendations

- Investigate why `http_req_failed` thresholds are absent from the test scripts - add them for completeness.
- Verify that sub-5ms responses reflect real computation and not mock/stub data in the test environment.
- The service is well-positioned for production - ensure the deployment environment matches the test environment.

7. Overall Performance Summary

The following table summarizes the production readiness of each service based on the combined results of all three test types. Readiness is assessed on response time stability, check pass rates, threshold compliance, and degradation behaviour under stress.

Service	Smoke	Avg Load	Stress	Peak RPS	Production Readiness
User Service	PASS	PASS	PASS	9.14/s	Ready with Monitoring
Compass Service	PASS	PASS	WARN	3.32/s	Not Ready - Optimize Now
Crystallization Service	PASS	PASS	PASS	14.43/s	Production Ready
Vision Service	PASS	WARN	FAIL	1.27/s	Not Ready - Major Work
Waste Valorization Service	PASS	PASS	PASS	15.59/s	Production Ready

7.1 Summary Findings

Based on the complete test suite, the following conclusions are drawn:

- Crystallization Service is the standout performer - sub-100ms responses even under 15 VUs with zero check failures across 3,112 stress requests. This service is immediately production-ready.
- Waste Valorization Service delivers exceptional sub-15ms response times at all load levels and is production-ready, though threshold definitions in test scripts should be completed.
- User Service is stable and production-capable with monitoring - the Get All Users endpoint requires caching to prevent latency spikes that appeared (11.8s max) under stress.
- Compass Service is not production-ready. Response times hit the 60-second timeout boundary during average and stress tests across multiple endpoints. Immediate database and query optimization is required.
- Vision Service is not production-ready for concurrent access. The heavy ML compute workload (Detections, Stats) causes 60-second timeouts under stress. Asynchronous processing and potential GPU acceleration are necessary.

7.2 Priority Action Plan

P	Service	Action Required	Impact
P1	Compass Service	Optimize Deals/Offers DB queries; add connection pooling, caching, and circuit breakers	Critical - prevents 60s timeouts
P1	Vision Service	Implement async job queue for Detections/Stats; explore ONNX/GPU acceleration	Critical - eliminates timeout failures
P2	User Service	Add Redis caching or pagination to Get All Users endpoint	High - prevents stress spikes
P3	Waste Valorization	Add http_req_failed thresholds to test scripts; validate non-mock data	Medium - test completeness

P	Service	Action Required	Impact
P3	Crystallization	Continue monitoring; no immediate action required	Low - already optimal

7.3 Next Steps

- Re-run tests after Compass and Vision Service optimizations to validate improvements.
- Establish continuous performance testing in the CI/CD pipeline using Grafana k6 cloud or self-hosted Grafana.
- Define SLA thresholds for each service (e.g., P95 < 500ms for User Service) and automate pass/fail gates.
- Gradually increase stress test VUs (30, 50, 100) as services are optimized to identify the true capacity ceiling.
- Complete threshold definitions in Waste Valorization test scripts.